

Final Year Project Proposal

**ContracthubPk: A Secure Web Platform for Managing Freelance Contracts
and Escrow Payments using Local Payment Gateways**

**Submitted to the
Department of Software Engineering
University Of Rasul, Mandi Bahaud Din**

**in Partial Fulfillment of
the Requirements for the Degree of
Bs Software Engineering
By**

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Project Proposal

Class: BS Software Engineering

Title: ContracthubPk: A Secure Web Platform for Managing Freelance Contracts and Escrow Payments using Local Payment Gateways

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Problem Statement: In Pakistan's freelance market, most of the time freelancers and clients rely on informal communication channels such as social media or messaging apps like WhatsApp, Facebook and LinkedIn to negotiate and manage projects. This often leads to trust issues, payment scam, lack of formal contracts, and project disagreements. Existing international platforms like Fiverr and Upwork charge high charges and do not support local payment gateways, making them less accessible for many users. So, there is a need for a secure, automated web platform that allows freelancers and clients to make digital contracts, manage escrow payments, and confirm transparent association using local payment methods such as Easypaisa and JazzCash.

Introduction: The proposed project with the titled "A Secure Web Platform for Managing Freelance Contracts and Escrow Payments using Local Payment Gateways," purposes to address the growing issues of trust, payment safety, and contract managing inside Pakistan's local freelance market. Many time freelancers and clients in Pakistan choose to work outside international platforms like Fiverr and Upwork due to their high package charges and strong competition. But, working independently over social media or informal communication medium often leads to payment disagreements, project delays, and a lack of responsibility, as there is no standardized system to confirm safe transactions or impose digital agreements.

This project proposes a web based platform that enables freelancers and clients to create, manage, and implement freelance contracts fast while confirming reasonable and transparent payments through escrow mechanisms. The system will integrate local payment gateways such as Easypaisa and JazzCash to support continuous and manageable financial transactions. Key features will include contract creation, milestone based payments, automated escrow release, disagreement resolve, and user confirmation to confirm consistency and confidence between both parties.

In automation of freelance contracts and confirming secure payment management, this system will help as a reliable digital bond between freelancers and clients in Pakistan. It will promote digital trust, financial presence, and transparent association, while reducing the dependence on expensive international freelancing platforms.

Purpose: The main purpose is to create a secure web platform that lets freelancers and clients to accomplish contracts and payments through an automatic escrow system. It purposes to remove trust problems and payment scam by integrating local gateways such as Easypaisa and JazzCash.

Scope: The scope of ContracthubPk is partially limited to design and implementation of a secure web based freelance contract and escrow management system for Pakistan's local market inside an academic environment.

Objectives:

- Joining freelancers and clients for transparent association.
- Applied an automatic escrow system to confirm safe and milestone based expenditures
- Integrate local payment gateways like Easypaisa, JazzCash
- Allow digital contract formation and managing for clarity
- Deliver a disagreement resolution and assessment system

Functional Requirements:

1. The system lets users to make an account and log in gradually as a **client** or **freelancer**.
2. Users can make and update their personal profiles, including basic info and payment details.
3. Both clients and freelancers can make a **digital agreement** by essential scope, cost, time and milestones.
4. The system allows users to view, accept, or reject contract appeals.
5. Clients can deposit contract payments using **local payment methods** such as Easypaisa or JazzCash.
6. The system holds deposited payments in an **escrow state** until promised conditions are satisfied.
7. Freelancers can submit completed work and request payment release.
8. Clients can assess submitted work and approve or request changes.
9. On client approval, the system automatically issues the escrow payment to the freelancer.
10. Users can raise disagreements if disagreements happen about work or payment.
11. The admin can review arguments, assess evidence, and decide payment consequences.
12. The system sends announcements to users for important activities such as contract creation, payment updates, and disagreement decisions.
13. Users can view their contract history, payment histories, and contract status.
14. The admin can manage users, contracts, payments, and system settings over a safe dashboard.
15. The system keeps logs of all major activities for transparency and examining.

Software / Tools needed:

- **Tools:**

Git and GitHub: GitHub offers a cloud based version control system that allows the entire team to collaborate on code, trail changes, and merge updates safely without overwriting each other's work

Jira: Jira supports Scrum and Kanban practices, helping the development team plan sprints, allocate tasks, and track development proficiently throughout the FYP lifecycle.

- **Technology:**

React Js

Component based, fast to build, large ecosystem (UI libraries, form handling). Next.js adds server side rendering and easy routing if you want SEO friendly pages (landing, docs).

Backend (ExpressJS + NodeJS): Lightweight, JavaScript end-to-end (same language as frontend), vast middleware ecosystem, good for building REST/GraphQL APIs.

PostgreSQL: PostgreSQL gives strong ACID guarantees, transactions, and relational integrity (useful for payments, contracts, audit logs)

Deployment Platform: ContractHubPk is deployed on free tier cloud platforms, with React applications hosted on Vercel, the backend on Render, and the database on MongoDB Atlas, making it cost effective and suitable for academic use.

Supervisor:

Signature: _____

